

Functional Specification Document (FSD)

1. Introduction

1.1 Purpose

The purpose of this Functional Specification Document is to outline the required features, system behavior, and functional requirements for the "Strategic Product Placement Analysis" visualization platform.

1.2 Scope

This system includes a data visualization pipeline utilizing Tableau for data processing and chart generation, and a Python Flask web application acting as the delivery mechanism for the embedded interactive dashboards.

2. Functional Requirements

2.1 Data Processing

- **FR1:** The system must import the `Product Positioning.csv` dataset.
- **FR2:** The system must accurately calculate `Price`, `Competitor Price`, and `Sales Volume`.

2.2 Visualizations

- **FR3:** The system must generate 8 distinct visualizations including Bar charts, Pie charts, and Bubble charts.
- **FR4:** The dashboard must include global interactive filters for `Demographics`, `Category`, and `Position`.
- **FR5:** A 3-scene Storyboard must be implemented to guide users through the specific insights regarding Clothing and Front of Store performance.

2.3 Web UI Application

- **FR6:** The Flask web server must render a Bootstrap-based frontend.
- **FR7:** The web UI must support embedded iframe `<script>` tags generated by Tableau Public.
- **FR8:** The web UI must provide a top navigation bar allowing users to jump between the Dashboard and Story views.

3. Non-Functional Requirements

- **Performance:** Dashboard load times within the web application must not exceed 5 seconds.
- **Usability:** The UI must adhere to a dark mode aesthetic with clear, intuitive navigation buttons.
- **Accessibility:** Ensure high contrast ratios between text and background elements.